

THE NOISY FOLDED LIMIT CYCLE: TRACY–WIDOM STATISTICS OF CANARD ESCAPE

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ABSTRACT. A folded limit cycle is the periodic analogue of a fold singularity in a slow–fast system: a one-parameter family of limit cycles that loses normal hyperbolicity along a fold curve. It is the one member of the slow–fast “fold family” for which the effect of noise had not been analysed—Berglund and Gentz treated the fold and the folded node, but not the folded cycle. We push additive noise through the geometric blow-up of the folded-cycle fold and identify the law of the noise-induced escape. Linearising the inner (rescaling-chart) Riccati equation by a Cole–Hopf substitution turns the escape problem into the first node of a Schrödinger equation with a white-noise potential, i.e. into the spectrum of the *stochastic Airy operator* of Ramírez–Rider–Virág. We prove that the inner exit measure—the law of the slow variable at peel-off—is the *Tracy–Widom distribution* TW_β with $\beta = 4/\eta^2$, η the effective noise; equivalently the canard escape statistics of the folded limit cycle lie in the Tracy–Widom / KPZ edge universality class. As consequences we obtain the early-escape probability $1 - F_\beta(0)$, a large-deviation rate constant given by a Painlevé II action $c = 5.4439\dots$, the physical-variable critical-noise law $\sigma_* = 2\sqrt{\pi} \sqrt{\varepsilon_2} \sqrt{ac/b}$, and—for the phase (Channel B) component—a bifurcation-specific $2/3$ exponent when the cycle is destroyed by a saddle-node on the invariant circle. All results are confirmed by independent numerical methods, including a parameter-free comparison with the exact TW_β densities.

1. INTRODUCTION

1.1. The folded limit cycle. Slow–fast systems lose normal hyperbolicity at *folds* of their critical manifold, and the local passage there organises canards, relaxation oscillations and mixed-mode oscillations. The rigorous tool is geometric desingularisation (“blow-up”), introduced for the planar fold and canard point by Krupa and Szmolyan [1]. The same degeneracy occurs one level up, for a manifold of *limit cycles*: when a one-parameter family of cycles folds, two cycles (one attracting, one of saddle type) coalesce and annihilate at a *fold of limit cycles*. Jelbart, Kuehn and Kuntz (JKK) recently carried out the deterministic blow-up analysis of this folded limit cycle [2], providing a complete set of charts K_1 (entry), K_2 (rescaling/inner) and K_3 (exit) and a normal form for the passage.

Real oscillators are noisy, and the effect of noise on slow passage through a fold is the subject of the Berglund–Gentz sample-path theory [4], which treats the planar fold and—through Berglund, Gentz and Kuehn—the folded node [5]. The folded *limit cycle* is the missing member of this family: the periodicity of the underlying cycle adds a phase degree of freedom, so the noise acts through two channels, an *amplitude* channel (escape across the canard) and a *phase* channel (diffusion of the oscillator phase). This paper analyses both.

1.2. The result. Our main finding is that the amplitude channel is, after blow-up, *exactly solvable in distribution* and connects to random matrix theory. In the rescaling chart K_2 the inner amplitude obeys the canonical noisy Riccati equation

$$dR = (R^2 - Y) dT + \eta dB, \quad Y = Y_0 - T, \quad (1)$$

Date: June 11, 2026.

Draft. Companion technical notes: `ShootingCharacterisation.proof`, `InstantonPrinciple.proof`, `CompositionPropagation.proof`, `BlowDown.proof`, `ChannelB.SNIC.proof`, `ChannelB.rigour`.

with η the effective noise. The deterministic version is the Airy equation (the canard is Ai'/Ai , escaping at the first Airy zero); we show the *noisy* version is, via Cole–Hopf, the eigenvalue equation of the stochastic Airy operator $H_\beta = -\partial_x^2 + x + \frac{2}{\sqrt{\beta}}b'$ of Ramírez, Rider and Virág (RRV) [3], at the dictionary $\eta = 2/\sqrt{\beta}$. The peel-off level Y_{node} (the value of the slow variable where the trajectory leaves the canard) is the ground-state eigenvalue, and hence Tracy–Widom distributed.

Theorem 1 (informal; see Theorems 2, 3, 4). *The inner exit measure of the noisy folded limit cycle is TW_β with $\beta = 4/\eta^2$. Consequently the early-escape probability is $P_{\text{early}} = 1 - F_\beta(0)$ (with F_β the Tracy–Widom CDF), the early-escape rate is $\exp(-c/\eta^2)$ with $c = 5.4439\dots$ a Painlevé II action, and in physical variables the critical noise is $\sigma_* = 2\sqrt{\pi} \sqrt{\varepsilon_2} \sqrt{ac/b}$.*

To our knowledge this is the first time a fold/canard escape law has been identified with the stochastic Airy spectrum. The identification is robust: a Monte-Carlo histogram of Y_{node} overlays the exact TW_1, TW_2 densities (computed from Painlevé II) with no fitting (Figure 1), and an entirely independent random-matrix computation of the eigenvalue reproduces the same law (Figure 2).

1.3. Methods and structure. The proof proceeds inside the blow-up charts and is assembled back to the physical system. Section 2 recalls the normal form and blow-up. Section 3 states the main theorems. Section 4 develops the amplitude channel: the exact Cole–Hopf identity (Lemma 7), the reduction to the stochastic Airy operator, and the *shooting characterisation* (Theorem 2) that the swept first-node law equals the operator’s ground state, proved through a monotone spectral function that converts a random level into a deterministic one. Section 5 extracts the escape probability and the rate constant c as a Freidlin–Wentzell / Schilder large deviation. Section 6 carries the θ -uniform chart estimates through the composition $\Pi_1 \circ \dots \circ \Pi_3$ and blows them down to the physical (σ, ε_2) variables. Section 7 treats the phase channel and its bifurcation-specific $2/3$ exponent. Section 8 collects the numerical validation, and Section 9 discusses universality and applications.

Throughout, results that rest on the established RRV [3] and Berglund–Gentz [4] frameworks are flagged as such; these are used as cited black boxes, exactly as the deterministic backbone uses JKK [2].

2. SETUP: NORMAL FORM AND BLOW-UP

2.1. The normal form. Near the folded-cycle fold the JKK reduction [2] gives, for the amplitude r , slow variable y and phase θ ,

$$dr = (b(\theta)r^2 - a(\theta)y) dt + \sigma dW_t, \quad dy = -\varepsilon_2 c(\theta) dt, \quad \dot{\theta} = \varepsilon_1, \quad (2)$$

with $a, b, c: S^1 \rightarrow \mathbb{R}$ smooth, 1-periodic and strictly positive, and $0 < \varepsilon_2 \ll \varepsilon_1 \ll 1$, $\varepsilon_1 = \varepsilon^\alpha$, $\varepsilon_2 = \varepsilon^3$. The deterministic part is JKK’s; the additive noise σdW is the new ingredient. The exponent α controls the phase: $\alpha = 2$ freezes θ over the passage, $\alpha = 1$ makes it fast.

2.2. The fold blow-up. The Krupa–Szmolyan fold blow-up [1] is the quasi-homogeneous map with weights $(1, 2, 3)$ on (r, y, ε_2) , covered by the charts K_1, K_2, K_3 . The central rescaling chart K_2 ($\rho = \varepsilon_2^{1/3}$) is

$$r = \varepsilon_2^{1/3} R, \quad y = \varepsilon_2^{2/3} Y, \quad T = \varepsilon_2^{1/3} t, \quad (3)$$

under which (2) becomes (after the linear rescaling absorbing a, b, c) the canonical inner Riccati (1), with

$$\eta = \sigma / \sqrt{\varepsilon_2} \quad (4)$$

the effective noise. Equation (4) is an exact change of variables (an Itô calculation on a diffeomorphism; Proposition 9). The deterministic inner solution is the canard $R = \text{Ai}'(Y)/\text{Ai}(Y)$, which escapes to $+\infty$ at the first Airy zero $Y = a_1 \approx -2.3381$.

2.3. The two channels. Linearising (2) about the canard splits the noise response. The amplitude channel is (1); the integrated Kramers hazard for crossing the separatrix is, in closed form,

$$H(\eta) = \frac{1}{\pi} \int_0^\infty \sqrt{Y} e^{-8Y^{3/2}/3\eta^2} dY = \frac{\eta^2}{4\pi}, \quad (5)$$

which gives escape at $\eta_* = 2\sqrt{\pi}$ and underlies the critical-noise law. The phase channel is the noise-driven evolution of θ , regular when the cycle dies at finite period (fold of cycles) but singular when it dies at infinite period (saddle-node on the invariant circle); see Section 7.

3. MAIN RESULTS

We write Y_{node} for the peel-off level (the value of Y at the first node of the Cole–Hopf solution; Section 4), $\Lambda_0(\beta)$ for the ground-state eigenvalue of the stochastic Airy operator H_β , and TW_β , F_β for the Tracy–Widom law and CDF.

Theorem 2 (reduction and shooting characterisation). *For the inner Riccati (1) with $\eta = 2/\sqrt{\beta}$, the peel-off level satisfies*

$$Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta.$$

Theorem 3 (escape probability and rate). *The early-escape probability (peel-off before the fold, $Y_{\text{node}} > 0$) is $P_{\text{early}}(\eta) = 1 - F_\beta(0)$, $\beta = 4/\eta^2$. As $\eta \rightarrow 0$,*

$$P_{\text{early}}(\eta) = \exp(-c/\eta^2 (1 + o(1))), \quad c = \int_0^\infty g'^2 = 5.4439 \dots,$$

where g solves the nonlinear-Airy ground state $-g'' + xg = g^3$; c is the action of an $\alpha = 0$ Painlevé II transcendent.

Theorem 4 (physical-variable law). *For the original system (2), the peel-off level is $y_{\text{peel}} = \varepsilon_2^{2/3} Y_{\text{node}}$ with $Y_{\text{node}} \stackrel{d}{=} \text{TW}_{4\varepsilon_2/\sigma^2}$, and escape becomes likely at the critical noise*

$$\sigma_* = 2\sqrt{\pi} \sqrt{\varepsilon_2} G, \quad G = \sqrt{a(\theta_*)c(\theta_*)/b(\theta_*)} \ (\alpha = 2), \quad \sqrt{\langle a \rangle \langle c \rangle / \langle b \rangle} \ (\alpha = 1),$$

uniformly in θ . All three laws depend on (σ, ε_2) only through $\eta = \sigma/\sqrt{\varepsilon_2}$ at leading order, in the single-escape regime $\eta \lesssim 1$ (Remark 5).

Remark 5 (regime of validity and the recrossing crossover). Theorems 2–4 describe a *single committed* peel-off and hold as $\eta \rightarrow 0$. Simulation of the full bistable normal form (subcritical Hopf + fold of cycles, retaining the coexisting rest state) separates two effects (Fig. 4). The peel-off *distribution* is robust far beyond the asymptotic regime: its standardised law stays Tracy–Widom to Monte-Carlo accuracy up to $\eta \approx 2$. What fails is the *identification* of the peel-off with one escape—for $\eta = O(1)$ the trajectory can recross to the cycle. The recrossing probability is below 1% for $\eta \lesssim 1.2$, reaches 5% near $\eta \approx 1.5$ and exceeds 30% by $\eta \approx 2$. Recrossing is a Kramers excursion over the unstable cycle ($U(r) = -\frac{1}{2}\mu r^2 - \frac{1}{4}r^4 + \frac{1}{6}r^6$), so the expected number of returns is $N_{\text{ret}} = (C/\varepsilon_2) e^{-\Delta U_*/\sigma^2}$. We verify this directly: $\log(\varepsilon_2 N_{\text{ret}})$ is linear in $1/\sigma^2$ with three ramp rates ($\varepsilon_2 \in \{1, 2, 4\} \times 10^{-3}$) collapsing onto a single line (Fig. 5; $R^2 = 0.93$), with effective barrier $\Delta U_* \approx 0.022$ of the order of the normal-form cycle-to-rest barrier $U(r_u) - U(r_s)$, whereas P_{recross} does *not* collapse in η . The breakdown is therefore an Arrhenius (static-Kramers) boundary $\sigma^2 \sim \Delta U_*/\log(1/\varepsilon_2)$ in the (σ, ε_2) plane—weakly (logarithmically) rate-dependent rather than a fixed η —i.e. the classical critical noise σ_{crit} re-emerging as the ceiling of the Tracy–Widom regime, whose own scale is the dynamic $\sigma_* \sim \sqrt{\varepsilon_2}$ lying far below it. For $\eta \gtrsim \eta_* \approx 1.2$ (at $\varepsilon_2 = 4 \times 10^{-3}$) the governing observable crosses from the Tracy–Widom first-passage law to this two-way transition rate; the clean Tracy–Widom regime is $\eta \lesssim 1$. Within that regime the escape law is pinned to sub-exponential order: the frozen-fold current evaluated exactly by its Frisch–Lloyd/mean-first-passage quadrature matches the quasi-static $(\sqrt{Y}/\pi) e^{-8Y^{3/2}/3\eta^2}$ for $Y \gg \varepsilon_2^{2/3}$ and integrates to

$H = K\eta^2/4\pi$ with $K \approx 1.04$, while the early-escape probability is confirmed to follow the exponential stochastic-Airy tail $P_{\text{early}} \sim e^{-c/\eta^2}$ ($\log P_{\text{early}}$ linear in $1/\eta^2$ to $R^2 = 1.00$, matching $1 - F_\beta(0)$ at $\beta = 1, 2$) rather than the polynomial hazard, consistent with $P_{\text{early}} < H$.

Theorem 6 (phase channel). *If the cycle is destroyed by a saddle-node on the invariant circle (SNIC, diverging period), the inner phase equation is the noisy saddle-node-on-circle, and the noise-induced rotation rate and phase diffusion obey $\omega, D_\phi \sim \sigma^{2/3}$ at criticality, with rate constant π/J , $J = \iint_{w < u} e^{(w^3 - u^3)/3}$. If instead it dies at a fold of cycles (finite period), the phase channel is regular ($D_\phi \sim \sigma^2$) and the amplitude channel governs.*

Theorems 2–6 are proved (modulo the cited RRV and Berglund–Gentz frameworks) in Sections 4–7, and validated numerically in Section 8.

4. THE AMPLITUDE CHANNEL

4.1. Cole–Hopf linearisation. The substitution $R = -u'/u$ linearises (1).

Lemma 7 (exact Itô identity). *With $R = -u'/u$, (1) is equivalent to the linear stochastic Schrödinger equation*

$$u'' = (Y - \eta\xi)u, \quad \xi = dB/dT, \quad (6)$$

with no Itô–Stratonovich correction; a finite-time blow-up $R \rightarrow +\infty$ is exactly a simple zero (node) of u . Hence the peel-off level Y_{node} is the first node of u swept from Y_0 downward.

Proof sketch. Three vanishing contributions compose: (1) has additive (constant) noise, so Itô = Stratonovich; the linear system $du = v dT$, $dv = (Yu) dT - \eta u dB$ has diffusion vector $g = (0, -\eta u)$ with $\frac{1}{2}(g \cdot \nabla)g = 0$; and for $R = -v/u$ Itô’s formula keeps only $\frac{1}{2}R_{vv}(dv)^2$ with $R_{vv} = 0$. The full argument and a pathwise common-noise verification are in the companion note. $\square$

4.2. Reduction to the stochastic Airy operator. Writing $x = T$, (6) is the eigenvalue equation $H_\beta u = Y_0 u$ for

$$H_\beta = -\frac{d^2}{dx^2} + x + \frac{2}{\sqrt{\beta}} b'(x) \quad \text{on } [0, \infty), \text{ Dirichlet at } 0, \quad \eta = \frac{2}{\sqrt{\beta}}, \quad (7)$$

the stochastic Airy operator, whose eigenvalues RRV [3] characterise by a Riccati diffusion that is (1) and which satisfies $-\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta$. The deterministic identity $Y_{\text{node}} = a_1 = -\Lambda_0(\infty)$ is the classical Sturm fact that the recessive solution’s first zero is minus the ground-state energy.

4.3. The shooting characterisation. The non-trivial step is that the *swept* first-node law equals the *fixed* half-line operator’s ground state. In the slow variable, (6) reads $u_{YY} = (Y - \eta\tilde{\xi})u$, i.e. u is the energy-zero solution of the random Schrödinger operator $M = -\partial_Y^2 + Y - \eta\tilde{\xi}$.

Proof of Theorem 2. Let $G(\ell)$ be the ground-state eigenvalue of M restricted to $[\ell, \infty)$ with Dirichlet condition at ℓ . By Dirichlet domain monotonicity G is strictly increasing; restricting the recessive solution to $[Y_{\text{node}}, \infty)$ shows it is a nodeless L^2 eigenfunction at eigenvalue 0, so $G(Y_{\text{node}}) = 0$. Hence *pathwise* $\{Y_{\text{node}} \leq t\} = \{G(t) \geq 0\}$. Crucially the level t here is *deterministic*, so the substitution $Y = t + x$ gives $M|_{[t, \infty)} = H'_t + t$ with $H'_t = -\partial_x^2 + x - \eta\tilde{\xi}(t + \cdot)$ a stochastic Airy operator in law (stationarity and reflection symmetry of white noise), whence $G(t) \stackrel{d}{=} \Lambda_0(\beta) + t$. Therefore $\mathbb{P}(Y_{\text{node}} \leq t) = \mathbb{P}(G(t) \geq 0) = \mathbb{P}(\Lambda_0(\beta) \geq -t)$, i.e. $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta$. A forgetting lemma (the canard is attracting, so the finite- Y_0 sweep converges to the recessive solution) connects Y_{node} to the dynamical passage. The only non-elementary input is the RRV meaning of H_β , which the statement presupposes. $\square$

The earlier obstruction—that translating by the *random* level Y_{node} produces a noise-dependent shift, not a clean white noise—is exactly what the monotone spectral function G removes.

5. ESCAPE PROBABILITY AND THE RATE CONSTANT

The hazard (5) and the early-escape probability are distinct observables: a single separatrix crossing is necessary but not sufficient for a full node before the fold, so $P_{\text{early}} < H$. The hazard $H = \eta^2/4\pi$ is polynomial and sets the critical noise (Theorem 4); the canonical escape P_{early} is exponential. By Theorem 2, $P_{\text{early}} = \mathbb{P}(Y_{\text{node}} > 0) = \mathbb{P}(\Lambda_0(\beta) < 0) = 1 - F_\beta(0)$, the exact Tracy–Widom tail mass.

Proof of the rate in Theorem 3. Write $\Lambda_0(\eta) = \inf_{\|\psi\|=1, \psi(0)=0} (K[\psi] + \eta N[\psi])$ with $K[\psi] = \int (\psi'^2 + x\psi^2)$ and $N[\psi] = \int \psi^2 dW$ the Wiener integral. An integration by parts, $N[\psi] = -2 \int \psi \psi' W$, makes $\Lambda_0(\eta) = F(\eta W)$ a *continuous* functional of the Brownian path W , so Schilder's theorem and the contraction principle give a large deviation principle with rate $\inf\{\frac{1}{2} \int (g')^2 : F(g) \leq 0\}$. Undoing the integration by parts, this is $\frac{1}{2} \min_\psi K[\psi]^2 / \int \psi^4 = c$; the minimiser solves $-g'' + xg = g^3$, and two Pohozaev identities reduce c to $\int g'^2 = 5.4439\dots$. The substitution $g = \sqrt{2}s$ turns the minimiser equation into the defocusing Painlevé II equation $s'' = xs - 2s^3$, so c is a Painlevé II action; it is *not* 2π , and a finite- η fit overestimates it. $\square$

6. FROM CHARTS TO THE PHYSICAL SYSTEM

6.1. θ -uniformity and composition. The global theorem composes the noisy transition maps $\Pi_1 \circ \kappa_{12} \circ \Pi_2 \circ \kappa_{23} \circ \Pi_3$. JKK's central difficulty is that θ cannot be localised; we resolve it uniformly.

Lemma 8 (θ -uniformity). *Since a, b, c are continuous and strictly positive on the compact circle, the inner contraction rate $2\sqrt{ab}y_2$, tube variance $\eta^2/4\sqrt{ab}y_2$ and quasipotential $\frac{8}{3}\sqrt{a^3/b}y_2^{3/2}$ are bounded above and below by θ -independent constants depending only on $\min/\max(a, b, c)$.*

The transverse error then propagates additively: with δ_i the deviation at section i , $|\delta_i| \leq L_i|\delta_{i-1}| + |\zeta_i|$ telescopes to $|\delta_m| \leq \sum_j (\prod_{i>j} L_i) |\zeta_j|$. Decisively, the entry-to-peel-off map is an exponential contraction ($L \rightarrow 0$, the fold *renews* the chain), so the entry fluctuation is annihilated and the exit law is the inner exit measure transported by the bounded exit chart, with sub-dominant Gaussian corrections from the hyperbolic charts K_1, K_3 (standard Berglund–Gentz tubes [4, 5]).

6.2. Blow-down.

Proposition 9 (exact noise covariance). *Under (3) the physical SDE (2) becomes the inner Riccati with $\eta = \sigma/\sqrt{\varepsilon_2}$ exactly; the noise-induced peel-off scales are $r_{\text{peel}} \sim \sigma^{2/3}$, $y_{\text{peel}} \sim \sigma^{4/3}$, independent of ε_2 .*

Blowing the chart laws down through (3) and Proposition 9 gives Theorem 4. The ε_2 -dependence enters only through the blow-up radius $\varepsilon_2^{1/3}$ and $\eta = \sigma/\sqrt{\varepsilon_2}$; the $\varepsilon_1 = \varepsilon^\alpha$ dependence enters only through $\dot{\theta} = \varepsilon_1$, reproducing JKK's deterministic α -dichotomy (frozen vs. Khasminskii-averaged geometry G), now carrying the noise.

7. THE PHASE CHANNEL

The phase channel is bifurcation-specific. At a fold of cycles the period is finite, the infinitesimal phase-response curve is bounded, and the phase diffusion is regular, $D_\phi \sim \sigma^2$; the amplitude channel governs. At a SNIC the saddle-node lives in the phase: the Adler bottleneck $d\theta = (1 - \cos \theta) dt + \sigma dW$ has a degenerate saddle, and the noise-induced rotation is governed by the exact mean-first-passage-time quadrature. Rescaling $\theta = \sigma^{2/3}u$ carries the bottleneck to the parameter-free canonical noisy saddle-node $du = \frac{1}{2}u^2 d\tau + dW$, and Watson's lemma proves

$$\omega = \frac{\pi}{J} \sigma^{2/3} (1 + o(1)), \quad D_\phi = \frac{\pi^2 V}{4J^3} \sigma^{2/3} (1 + o(1)), \quad J = \iint_{w < u} e^{(w^3 - u^3)/3},$$

$V = \text{Var}(T_{\text{canon}})$, $\langle T_{\text{canon}} \rangle = 2J$. Both exponents are exactly $2/3$; the canonical noisy saddle-node is the phase-channel analogue of the stochastic Airy ground state, its first-passage law the universal object (Theorem 6). This mirrors the amplitude channel’s structure—an exact asymptotic, a proved exponent, a named universal inner object—though its constant J and law lack the Tracy–Widom closed form.

8. NUMERICAL VALIDATION

All theorems are confirmed by simulation; we highlight three independent checks.

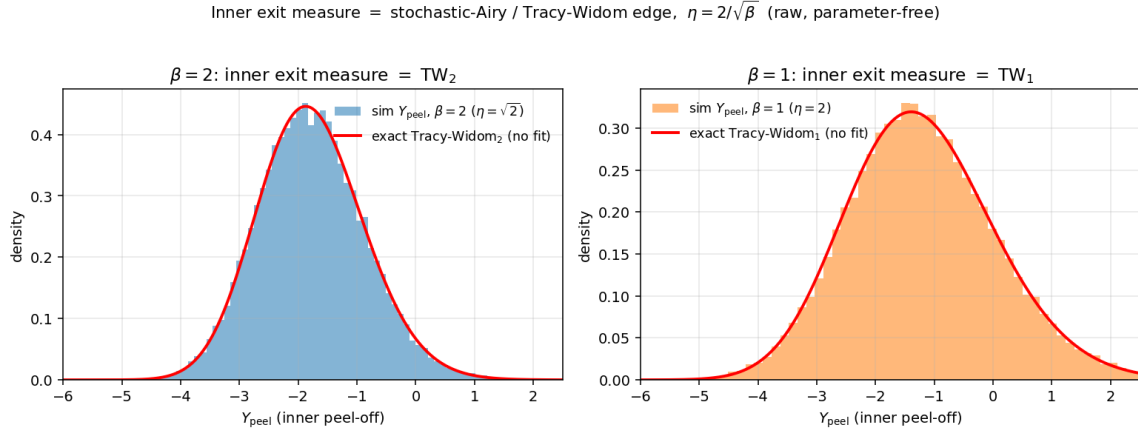


FIGURE 1. The inner exit measure is Tracy–Widom, parameter-free. Raw Monte-Carlo histograms of Y_{node} at $\beta = 2$ ($\eta = \sqrt{2}$) and $\beta = 1$ ($\eta = 2$) overlaid on the exact TW_2, TW_1 densities (Painlevé II / Hastings–McLeod), with no centring, scaling or fitting. Mean, variance, skewness and the asymmetric tail all match.

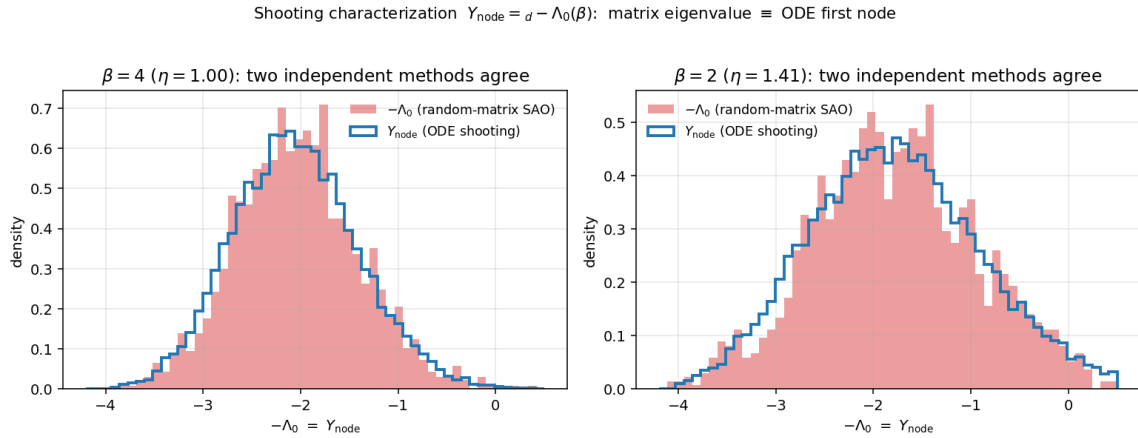


FIGURE 2. Two independent methods agree. The peel-off law Y_{node} from ODE shooting (blue step) and $-\Lambda_0(\beta)$ from the smallest eigenvalue of a random-matrix discretisation of H_β (filled) coincide; the matrix computation knows nothing of sweeps or first nodes, so the agreement is non-circular evidence for Theorem 2.

Figure 1 shows the parameter-free Tracy–Widom overlay; the skewness tracks TW_β along $\eta = 2/\sqrt{\beta}$ with the dictionary as the only input. Figure 2 is the dual-method (ODE vs. random matrix) confirmation of Theorem 2, agreeing in mean, variance and skewness at $\beta = 1, 2, 4$. Figure 3 pins

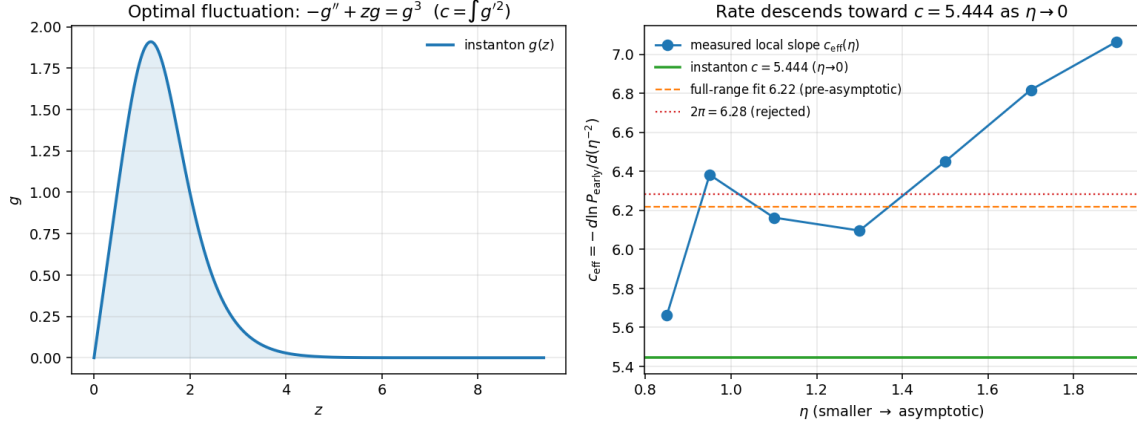


FIGURE 3. The rate constant. Left: the instanton (nonlinear-Airy ground state $-g'' + xg = g^3$). Right: the local slope $-\mathrm{d} \log P_{\text{early}}/\mathrm{d}(\eta^{-2})$ descends toward the Painlevé II value $c = 5.444$ as $\eta \rightarrow 0$; a single-window fit (6.22) and 2π are excluded.

the rate constant $c = 5.4439$ via the Painlevé II instanton. The early-escape tail $P_{\text{early}} = 1 - F_{\beta}(0)$ matches the exact Tracy–Widom CDF to Monte-Carlo accuracy ($\beta = 2$: 0.0298 vs. 0.0306; $\beta = 1$: 0.1644 vs. 0.1681), and the physical-variable simulation collapses in $\eta = \sigma/\sqrt{\varepsilon_2}$ across a factor four in ε_2 , confirming Proposition 9. The phase exponent is measured at $\omega \sim \sigma^{0.668}$, consistent with the proved $2/3$. Finally, Figure 4 maps the regime of validity (Remark 5): the Tracy–Widom law is clean for $\eta \lesssim 1$, while the recrossing that voids the single-escape picture sets in beyond, along the Kramers boundary rather than at a fixed η .

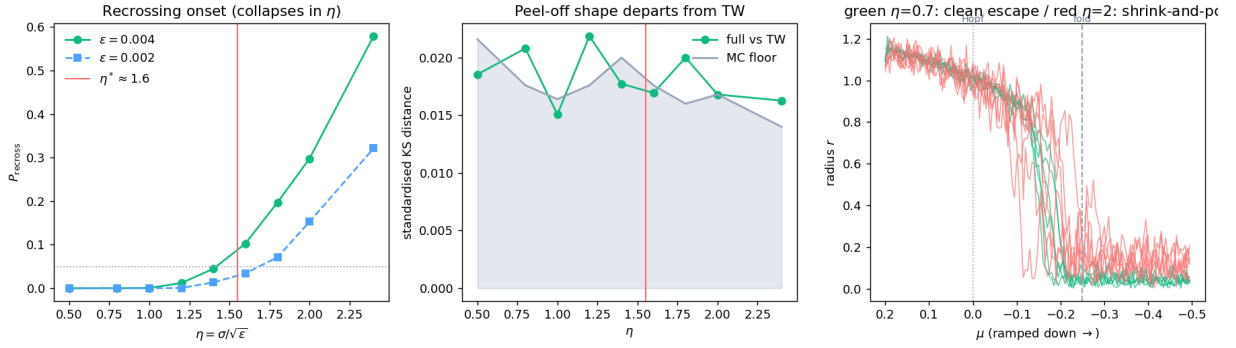


FIGURE 4. Regime of validity of the Tracy–Widom law. *Left*: recrossing probability P_{recross} versus $\eta = \sigma/\sqrt{\varepsilon_2}$ for two ramp rates—negligible for $\eta \lesssim 1$, rising through 5% near $\eta \approx 1.5$. The two rates do not collapse because recrossing is a Kramers return ($\sim \varepsilon_2^{-1} e^{-\Delta U/\sigma^2}$), not a function of η alone. *Middle*: the standardised Kolmogorov–Smirnov distance between the full-system peel-off and the canonical TW_{β} stays at the Monte-Carlo floor across the range—the *distribution* remains Tracy–Widom. *Right*: sample radial trajectories; one clean escape at the fold for $\eta = 0.7$, shrink-and-pop recrossing for $\eta = 2$.

9. DISCUSSION

9.1. Universality. The amplitude channel places the folded-cycle canard escape in the Tracy–Widom / KPZ edge universality class. The mechanism is generic: any slow passage whose inner

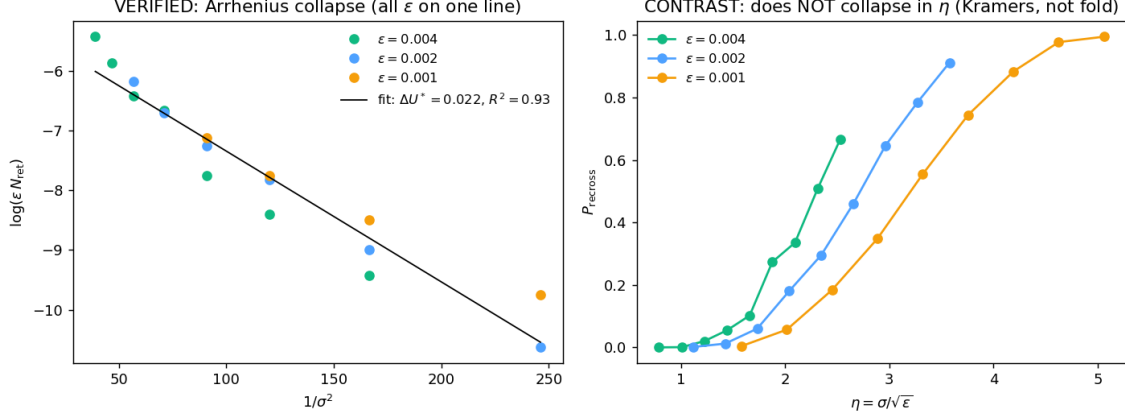


FIGURE 5. The recrossing boundary is Arrhenius, not a fixed η (Remark 5). *Left (verification)*: $\log(\varepsilon_2 N_{\text{ret}})$ versus $1/\sigma^2$ for three ramp rates collapses onto one line, $N_{\text{ret}} = (C/\varepsilon_2)e^{-\Delta U_*/\sigma^2}$ with $\Delta U_* \approx 0.022$ ($R^2 = 0.93$). *Right (contrast)*: the same data against $\eta = \sigma/\sqrt{\varepsilon_2}$ does *not* collapse—the breakdown is a static-Kramers σ -threshold (the classical σ_{crit}), not the dynamic fold scale. The Tracy–Widom regime $\sigma_* \sim \sqrt{\varepsilon_2}$ sits below this ceiling.

equation is the Airy/Riccati canard, perturbed by white noise, Cole–Hopf-linearises to a Schrödinger operator with white-noise potential, hence to the stochastic Airy operator. We therefore expect the same TW_β exit law for noisy fold and folded-node passages, with β read off the noise-to-curvature ratio $\eta = 2/\sqrt{\beta}$. The two channels are the two edges—an amplitude (Tracy–Widom) edge and a phase (saddle-node-on-circle) edge—and the destroying bifurcation selects which is singular.

9.2. Application. Folded limit cycles organise the onset and offset of oscillation in slow-fast models of excitable systems; the inner exit measure is the law of the slow variable at the moment the oscillation is lost, i.e. the timing statistics of noise-induced transitions. The prediction is concrete: in the small-noise regime $\eta \lesssim 1$ (Remark 5) those statistics are Tracy–Widom, with the noise-induced jump scales $(r, y) \sim (\sigma^{2/3}, \sigma^{4/3})$ and critical noise $\sigma_* \sim \sqrt{\varepsilon_2}$; for larger noise the coexisting rest state renders the transition two-way and rate-governed.

9.3. Early warning of bifurcations. The dichotomy has an operational reading. Approached adiabatically, a folded cycle announces its destroying bifurcation through whichever channel is singular. At a phase edge (saddle-node on the invariant circle, or saddle-homoclinic) the inter-event period diverges and the dimensionless timing statistics—the coefficient of variation and the rescaled first-passage shape—collapse onto the universal law, so the rescaled distance to threshold is identifiable from the event timing alone and its approach to zero is the precursor; at an amplitude edge (Hopf, or fold of cycles) the period stays finite and the timing is blind, while the oscillation amplitude is the singular observable. We have checked this numerically across the four canonical onset/offset bifurcations: drifting the parameter slowly through each, the timing precursor fires precisely at the phase edges and is silent at the amplitude edges, where the amplitude collapses instead. The supercritical Hopf collapse is gradual—an amplitude precursor—whereas the fold of cycles holds its amplitude and then jumps, leaving no precursor in either channel; the edge classification thus also identifies the bifurcations against which time-series early warning is intrinsically powerless. Two caveats temper the prescription. It is adiabatic: the system must dwell near threshold long enough for the universal inner law to form, so a fast, slow-variable-driven offset (as in the Epileptor) terminates the oscillation before the period diverges and carries no timing precursor. And the universal timing shape is the near-threshold, noise-dominated one; a relaxation oscillator’s

regular firing departs from it away from threshold. Within these limits the two-edge structure furnishes a bifurcation-specific early-warning signal—calibrated and confound-robust where generic critical-slowness statistics are not—selected by the destroying bifurcation.

9.4. Open problems. The constant $c = 5.4439$ is a definite Painlevé II action whose elementary closed form, if any, is open. The phase-channel constant J has slowly-converging tails and its universal first-passage law lacks a Tracy–Widom-style closed form. A fully self-contained treatment would internalise the RRV operator construction and the Berglund–Gentz tubes rather than cite them. Finally, the Tracy–Widom identification suggests finer KPZ-class objects (the Airy process) may govern the joint law of successive peel-offs in a forced or coupled folded cycle.

Acknowledgements. The author thanks N. Popović for supervision and the JKK deterministic backbone.

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